



1.The SDK27 series integrated single-phase voltage regulation module integrates a phase-shifting trigger circuit, unidirectional or bidirectional thyristors, RC impedance capacitance absorption circuit, and power supply circuit. It can automatically or manually adjust the voltage on the load to regulate the output power.

2.Fully compatible inputs such as 0-5VDC, 0-10VDC, 4-20mA, etc Automatic control mode, also available for manual control, output voltage from Linearly adjustable from 0V to maximum value.

3.The module is equipped with a built-in power transformer to form an integrated structure,Simple wiring, easy to use, and extremely cost-effective.

4. The module adopts SMT technology and DCB ceramic substrate, with stable performance and high reliability, and can adapt to inductive or resistive loads.

5. The module has LED power indicator and LED output adjustment indicator.

6. The input control terminals are designed with full isolation from the strong current main circuit, and the insulation medium withstand voltage is greater than 2000 Vac.

SELECTION GUIDE

load	type	Selection of Solid State Relays
Pure resistance (small change in cold and thermal resistance)	● NiCr ● FeCr ● FeAlCo	Actual load current=nominal value of solid-state relay/3
inductive	● transformer ● coil ● fan ● water pump ● motor	Actual load current=nominal value of solid-state relay/5

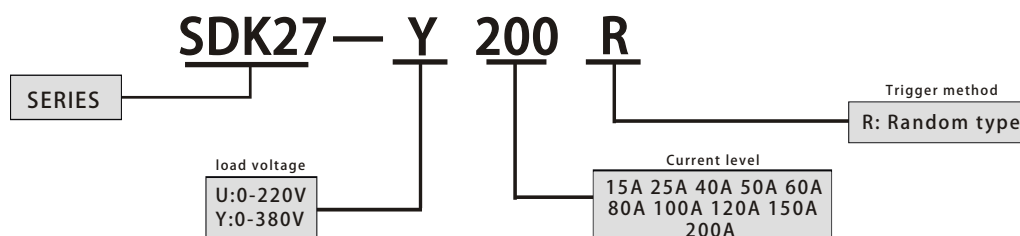
■ Built in RC protection

■ LED indicator

■ Work adaptation temperature: -30-40 °C

■ The bottom plate is made of high conductivity oxygen free copper

■ The shell is made of UL94-V0 engineering flame-retardant plastic (PBT) material



INSTALLATION INSTRUCTIONS

- The surface of the radiator should be smooth and free of burrs; According to the electrical safety standards for distribution cabinets, it is not recommended to ground the radiator to avoid high-energy arcing caused by short circuits.
- The contact surface between the base plate of the high-power relay and the radiator should be padded with our SDK-10 thermal pad, and the relay and radiator should be fastened with standard screws.
- For the SSR of DCB ceramic base plate, be careful to handle it with care to avoid dropping and collision. During installation, apply even force to prevent the ceramic base plate from breaking.
- The cross-sectional area of power wires can be selected from 4 to 8A per square millimeter, and flexible wire transitions are generally used for wiring. The lead out of the wire should have a support to avoid terminal breakage under stress.
- Installation environment requirements: It should be installed vertically on a well ventilated chassis, and attention should be paid to fully utilizing the heat dissipation conditions of air convection;
If it is a high temperature and high humidity environment, the joint should also be coated with conductive paste to reduce heat generation, prevent moisture, electrical and chemical corrosion.

■ Safe operation:

Maintenance personnel must first cut off the AC power supply before checking the output circuit, as the output terminals are live.

■ Common applications

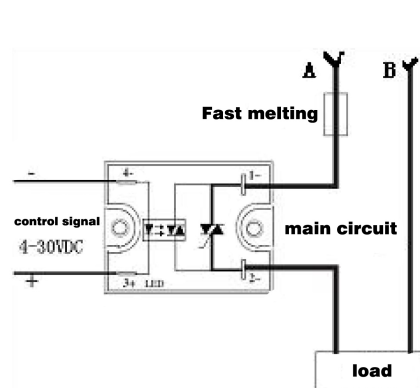


Figure 1: Application in single-phase electricity

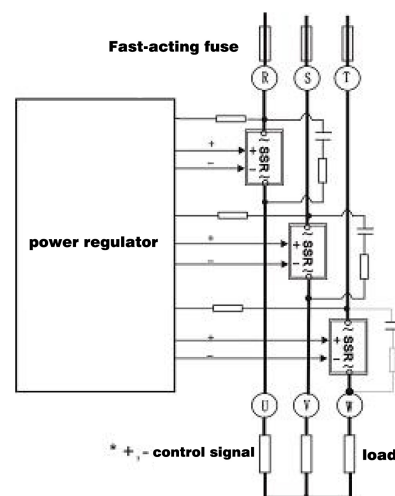


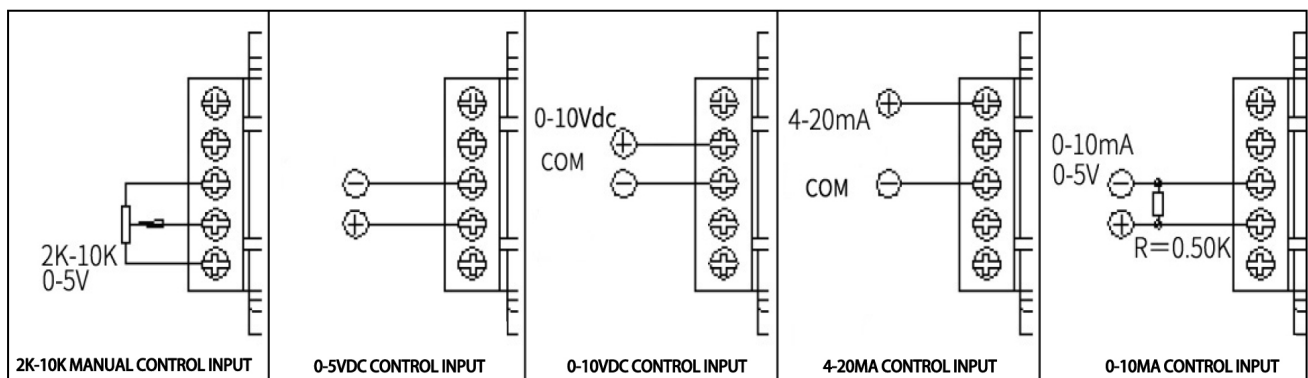
Figure 2: Application in three-phase electricity

Installation environment requirements:

It should be installed vertically on a well ventilated chassis, and attention should be paid to fully utilizing the heat dissipation conditions of air convection; If it is a high temperature and high humidity environment, the joint should also be coated with conductive paste to reduce heat generation, prevent moisture, electrical and chemical corrosion.

WIRING INSTRUCTIONS(INPUT)

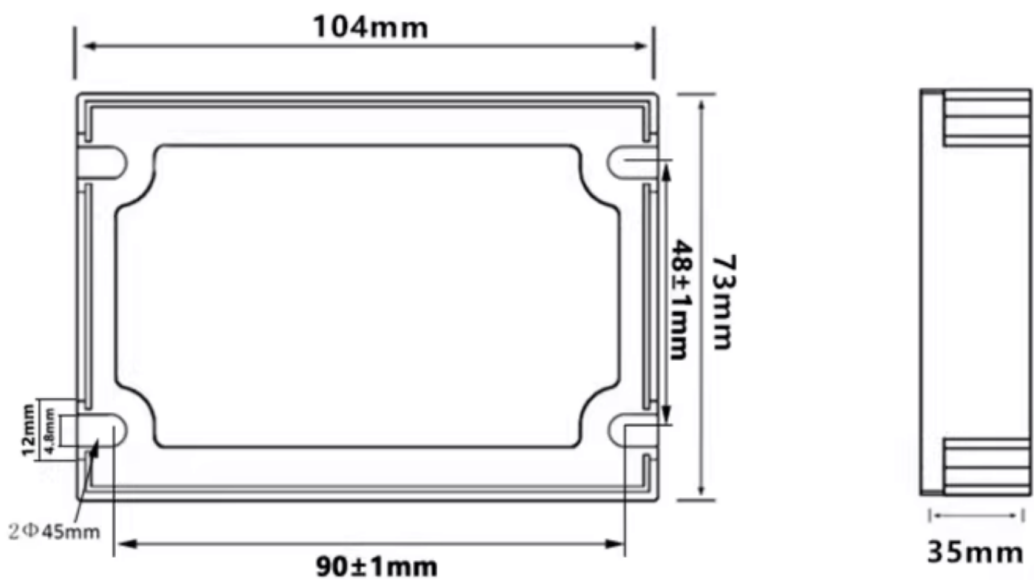
The unique fully compatible input control mode can adapt to automatic modes such as 0-5VDC, 0-10VDC, 4-20mA, 1-5VDC, and manual potentiometers, without the need for special customization. It can also be manually controlled by potentiometers. Wide input adjustment range, high output adjustment accuracy, and strong anti-interference ability.



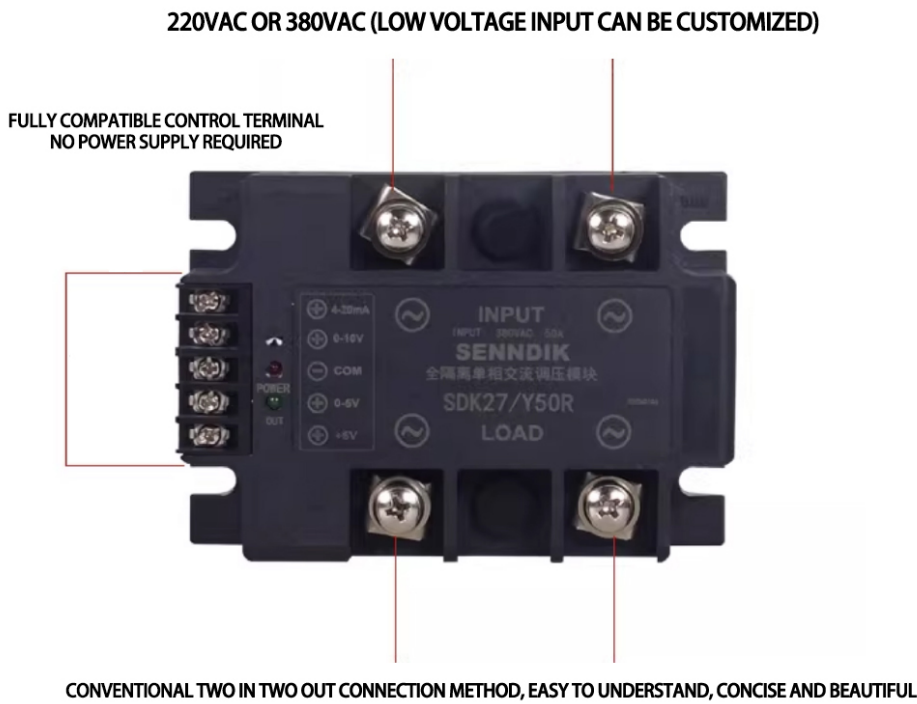
INSTRUCTIONS FOR USING INPUT TERMINALS

1. Manual control mode of potentiometer: According to the diagram, the middle end of the potentiometer is connected to the 0-5V terminal of the module, and the other two ends of the potentiometer are respectively connected to the COM terminal and +5V terminal of the module. +The 5V voltage is generated internally by the module itself and does not require external supply. It is only used in conjunction with a manual potentiometer and is not used for other purposes. The selected potential. The resistance value of the device is between 2-10K Ω . When the control terminal cont changes from 0-5VDC, the voltage on the AC load is linearly adjustable from 0 volts to the maximum value. The higher the voltage at the 0-5V terminal, the greater the module output.
2. 0-5VDC control mode: As shown in the diagram, it can accept 0-5VDC analog signals from microcontrollers and other devices. The positive input is connected to the 0-5V terminal, and the negative input is connected to the COM terminal. The input impedance of the 0-5V terminal relative to the COM terminal inside the module is greater than 30K Ω . When the control terminal 0-5V changes from 0-5VDC, the voltage on the AC load can be linearly adjusted from 0 volts to the maximum value. When cont is around 0-0.7Vdc, it is a fully closed area, reliably turning off the output of the entire circuit; The adjustable range of 0-5V is around 0.7Vdc-4.3Vdc, which means that as the control voltage increases, the phase shift angle α linearly decreases from 180° to 0° , the conduction angle increases, and the voltage on the AC load increases from 0 volts to the maximum value; When 0-5V is around 4.3Vdc-5Vdc, it is the fully open area, and the voltage on the AC load is at its maximum value (close to the grid voltage).
3. 0-10VDC control mode: As shown in the diagram, it can accept 0-10VDC analog signals from PLC, etc. The input impedance between the 0-10VDC end and the COM end inside the module is greater than 15K Ω .
4. 4-20mA control mode: As shown in the diagram, it can accept 4-20mA analog signals from temperature controllers, etc. The input impedance of the 4-20mA end relative to the COM end inside the module is 250 Ω . When controlling the input with 4-20mA, the fully closed area is around 4-5mA, which reliably shuts off the output of the entire circuit; The adjustable range is around 5-19mA, which means that as the control current increases, the phase shift angle α decreases linearly from 180° to 0° , and the voltage on the AC load increases from 0 volts to the maximum value; When it is around 19-20mA, it is the fully open area and the output is at its maximum.
5. 0-10mA control mode: According to the diagram, when using this mode, it is necessary to connect the 0-5V end of the module with
Connect a 500 Ω , 1/2W resistor between the com terminals, and when 0mA is input, the 0-5V terminals are 0Vdc, When inputting 10mA, the 0-5V terminal is 5VDC.
6. Each functional end must be positive relative to the com end, and the com end must be negative. If the polarity is reversed, the module will be
The output terminal of the block main circuit may lose control.
7. The control characteristics of each functional end of the module are all positive, that is, the higher the control voltage, the stronger the module
The higher the output voltage of the main circuit.
8. At a certain moment, it is advisable to use one input control method. If two or more methods are input simultaneously
When used, it usually plays a major role with a stronger input signal. Modules can be manually and Automatic dual-use, such as automatic connection to 4-20mA terminal and manual connection to 0-5V terminal, can be achieved through Double throw switch for functional switching.

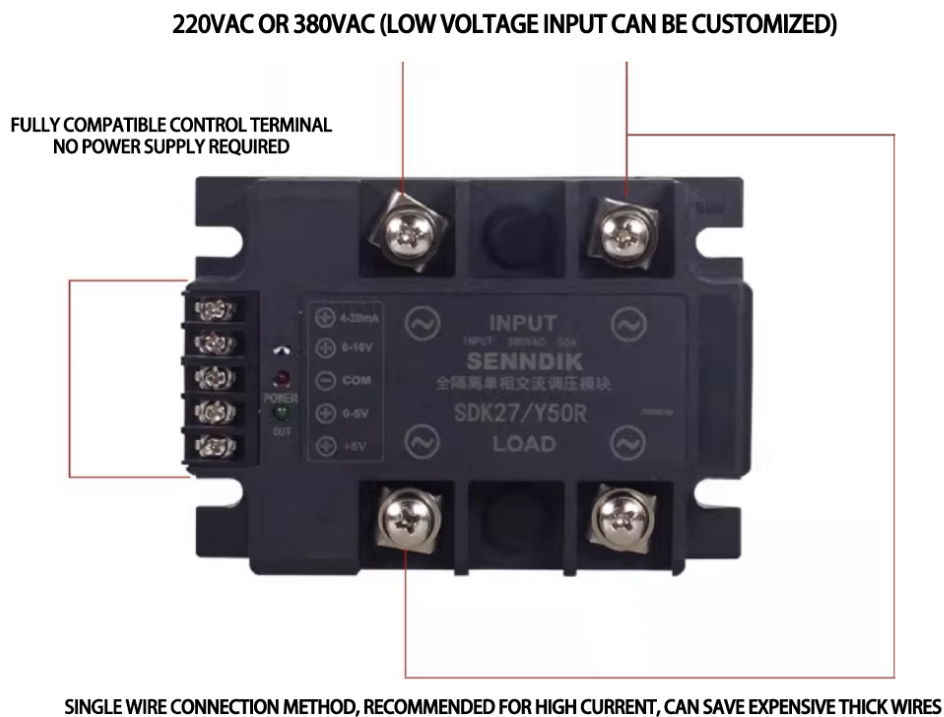
OVERALL DIMENSIONS



ELECTRIC WIRING DIAGRAM



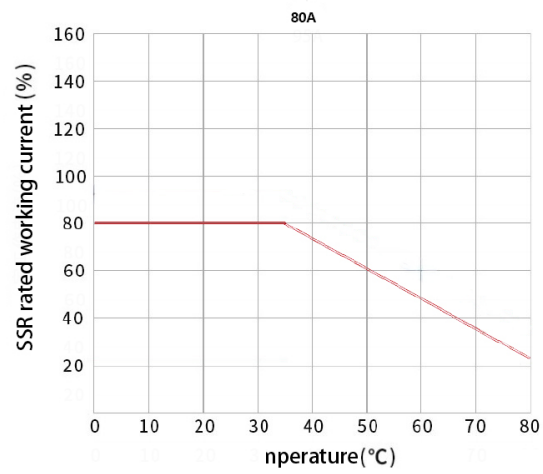
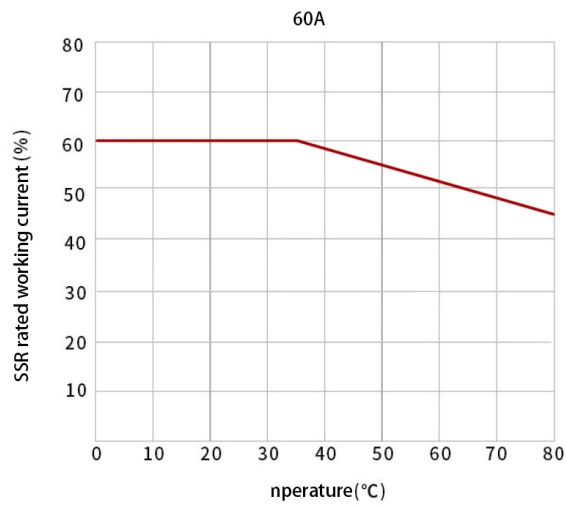
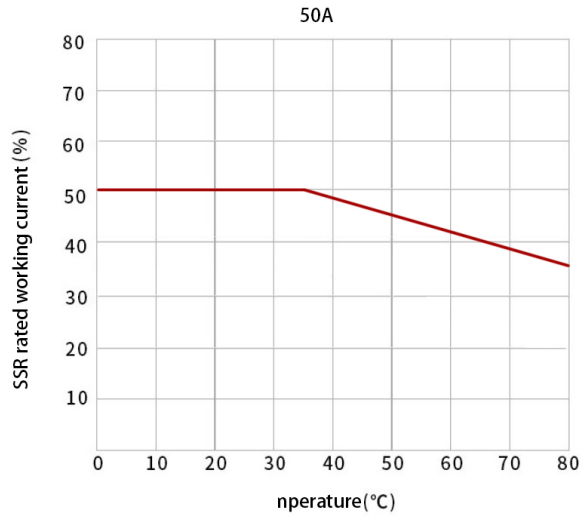
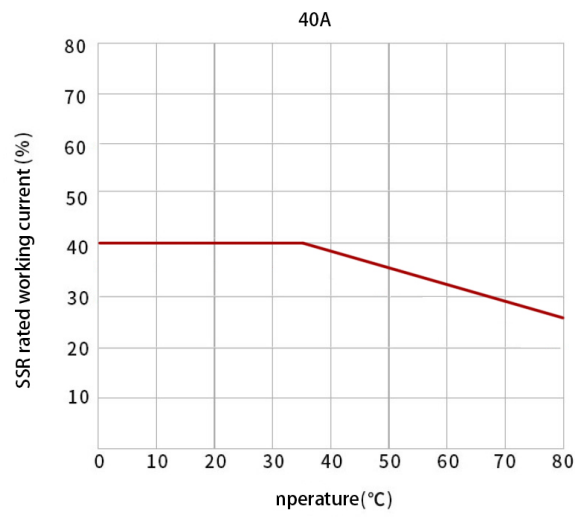
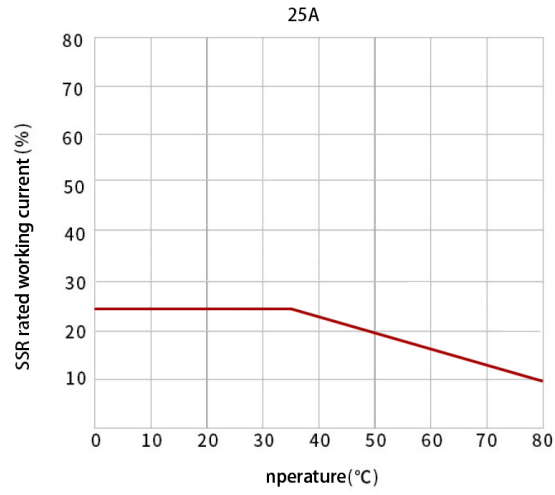
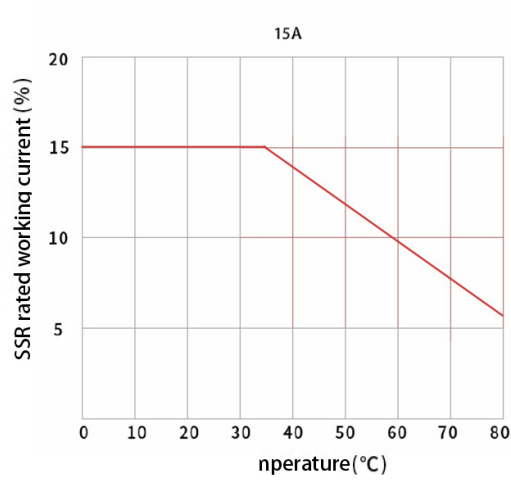
ELECTRIC WIRING DIAGRAM



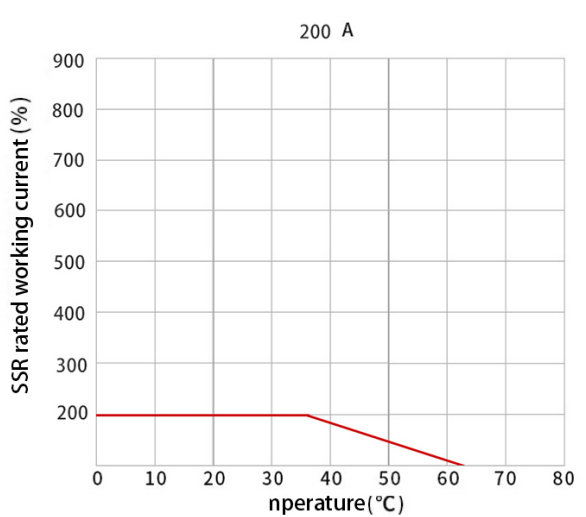
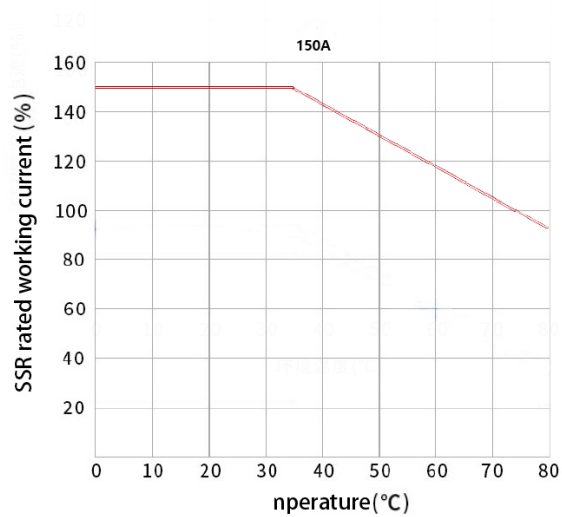
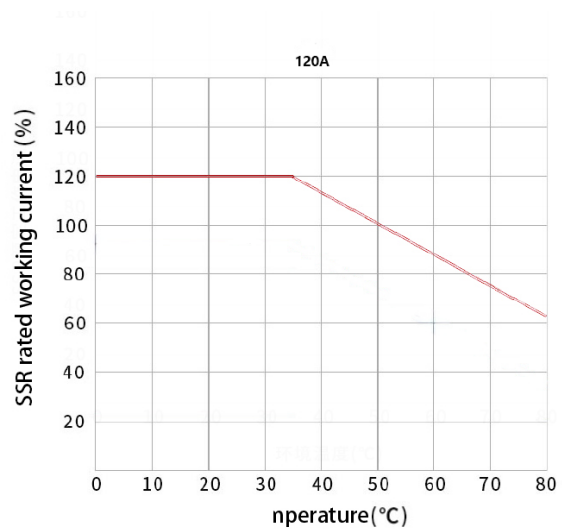
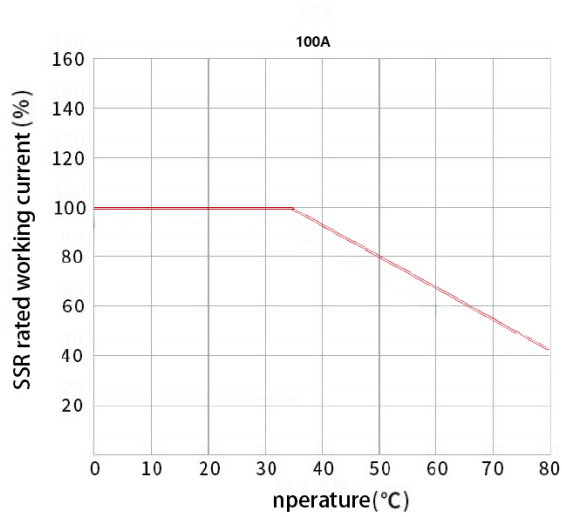
Note: It is generally recommended to use double wire connection for small currents below 70A specification, and for large currents above 90A specification

Use single wire connection method for grid connection.

TEMPERATURE



TEMPERATURE



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若有更改, 恕不另行通知

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